# BLOWING UP THE IDEAL OF DENOMINATORS OF A REGULAR MEROMORPHIC SECTION MAKES THAT SECTION A GLOBAL REGULAR SECTION MODULO THE EXCEPTIONAL DIVISOR

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 9232-9248 and proof span 9250-9286 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The proof refers to two definitions of the chapter (the denominator ideal and the invertible sheaf of a divisor difference); both references are part of the counted span and are handled as navigation pointers.

*Editorial note.* The closing parenthesis giving the divisor cut out by the numerator is the author's and is preserved.

*Editorial note.* 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- The definition of the ideal of denominators of a regular meromorphic section and its properties, cited from the same chapter. Cited at proof lines 9250-9252. Gives admissibility of the blowing up and lets the section be written locally as a quotient of nonzerodivisors.
- The notation for the invertible sheaf associated with an effective Cartier divisor and for the twisting of a line bundle by the difference of two divisors, cited from the same chapter. Cited at proof lines 9253-9256. Makes the statement meaningful and converts it into a statement about a regular section of a twisted bundle.
- Pullbacks of meromorphic sections along a blowing up and along a morphism with integral fibres, cited from the same chapter. Cited at proof lines 9257-9260. Shows that the pullback of the given meromorphic section is defined and nonvanishing in the appropriate sense.

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- The affine local description of a blowing up by blowup algebras and the identification of the exceptional divisor over each chart with the divisor cut out by the chosen element. Cited at proof lines 9272-9276. Supplies the charts and the generator of the twist used in the local computation.

*Source reference.* Original definition 01WX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2498–2498. Same-chapter reference used in the retained author text at source line(s) 3576, 6928, 9257. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original definition 02P1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3536–3536. Same-chapter reference used in the retained author text at source line(s) 9253. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 0809: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10174–10174. Same-chapter reference used in the retained author text at source line(s) 8602, 9259. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 0804: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10169–10169. Same-chapter reference used in the retained author text at source line(s) 8406, 8524, 8536, 8578, 8731, 8837, 8956, 9203, 9275. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 02OY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3533–3533. Same-chapter reference used in the retained author text at source line(s) 9260. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $X$ be a locally Noetherian scheme. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $s$ be a regular meromorphic section of $\mathcal{L}$. Let $U \subset X$ be the maximal open subscheme such that $s$ corresponds to a section of $\mathcal{L}$ over $U$. The blowup $b : X' \to X$ in the ideal of denominators of $s$ is $U$-admissible. There exists an effective Cartier divisor $D \subset X'$ and an isomorphism*

$$b^*\mathcal{L} = \mathcal{O}_{X'}(D - E),$$

*where $E \subset X'$ is the exceptional divisor such that the meromorphic section $b^*s$ corresponds, via the isomorphism, to the meromorphic section $1_D \otimes (1_E)^{-1}$.*

*Proof.* From the definition of the ideal of denominators in Definition 02P1 we immediately see that $b$ is a $U$-admissible blowup. For the notation $1_D$, $1_E$, and $\mathcal{O}_{X'}(D - E)$ please see Definition 01WX. The pullback $b^*s$ is defined by Lemmas 0809 and 02OY. Thus the statement of the lemma makes sense. We can reinterpret the final assertion as saying that $b^*s$ is a global regular section of $b^*\mathcal{L}(E)$ whose

zero scheme is $D$. This uniquely defines $D$ hence to prove the lemma we may work affine locally on $X$ and $X'$. Assume $X = \mathrm{Spec}(A)$ is affine and $\mathcal{L} = \mathcal{O}_X$. Then $s$ is a regular meromorphic function and shrinking further we may assume $s = a'/a$ with $a', a \in A$ nonzerodivisors. Then the ideal of denominators of $s$ corresponds to the ideal $I = \{x \in A \mid xa' \in aA\}$. Recall that $X'$ is covered by spectra of affine blowup algebras $A' = A[\frac{I}{x}]$ with $x \in I$ (Lemma 0804). Fix $x \in I$ and write $xa' = aa''$ for some $a'' \in A$. The divisor $E \subset X'$ is cut out by $x \in A'$ over the spectrum of $A'$ and hence $1/x$ is a generator of $\mathcal{O}_{X'}(E)$ over $\mathrm{Spec}(A')$. Finally, in the total quotient ring of $A'$ we have $a'/a = a''/x$. Hence $b^*s = a'/a$ restricts to a regular section of $\mathcal{O}_{X'}(E)$ which is over $\mathrm{Spec}(A')$ given by $a''/x$. This finishes the proof. (The divisor $D \cap \mathrm{Spec}(A')$ is cut out by the image of $a''$ in $A'$.) $\square$